

Robot Scientist Worksheet

Help Doodlebot rescue Princess-Bot, and find out how robots find their way.



Name: **Partner:** **Date:**

1 Predict

What if Doodlebot's sensor could see **4 squares** instead of **1**? Circle your guesses.

A. Steps to finish mapping will be…

◯ **more** ◯ **fewer** ◯ **the same**

B. The shortest path to Princess-Bot will be…

◯ **longer** ◯ **shorter** ◯ **the same**

2 Test

Maze ID 1X6Y (Lesson maze)

Keep the same: maze, Doodlebot, Princess-Bot, BFS. **Change only:** the sensor range.

	Run 1: sensor range 1	**Run 2: sensor range 4**
Steps to finish mapping how many moves Doodlebot made while exploring		
Shortest path length moves on the green rescue path to Princess-Bot		

Was my prediction right? A: ☐ yes ☐ no

When the sensor range increased, steps to finish mapping _____,
and the shortest path _____.

Challenge (if you finish early): sensor 2 = _____ steps, sensor 3 = _____ steps. What do you notice?

3 Reflect

1. Which run finished mapping in fewer steps? Why?

2. Was the shortest path different? Why or why not?

3. Think of a real robot. What are its "black squares", the things it can't know until it senses them?

How Doodlebot finds its way: **explore** (sense → map → move, again and again) → back to start → **plan** the shortest path → **rescue**